TARIFUL HOQUE



CSE Graduate | Machine Learning & AI | Data Science | UI/UX

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Portfolio: myportfoloiwebsite.netlify.app

PROFESSIONAL SUMMARY

Computer Science and Engineering graduate from East West University and IT Intern in the AI Integration & Cybersecurity wing at YSSE. I build end-to-end machine learning and AI systems, from data collection and model training to deployment. Built an IoT air-quality model (Random Forest, R^2 0.9933 on a time-ordered hold-out) shipped as a live web app, and a grounded RAG service with FastAPI, FAISS and the Claude API. Skilled in Python, JavaScript, data analysis and UI/UX design, with 5 live web applications deployed. Seeking an entry-level AI/ML Engineer, Data Scientist or Data Analyst role.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript (ES6+), C, SQL, HTML5, CSS3
- **Machine Learning & AI:** scikit-learn, Random Forest, LightGBM, KNN, SVM, feature engineering, model evaluation, Retrieval-Augmented Generation (RAG), embeddings, FAISS vector search, LLM guardrails, prompt engineering, Claude API
- **Data:** data analysis, data preprocessing, data visualization, IoT sensor data, Chart.js, MS Excel
- **Web & Backend:** FastAPI, REST APIs, Progressive Web Apps (PWA), service workers, Tailwind CSS, responsive and accessible UI
- **Tools & DevOps:** Git, GitHub, GitHub Actions (CI), Docker, pytest, Netlify, GitHub Pages, Cloudflare
- **Design:** UI/UX design, Figma, product prototyping, bilingual (Bangla / English) interfaces
- **Professional:** presentation, technical writing, team collaboration, leadership, time management, fast learning

EXPERIENCE

IT Intern, AI Integration & Cybersecurity Wing
Youth School for Social Entrepreneurs (YSSE) | Sep 2026 – Present

- Prepare and present the wing's weekly update deck at the official departmental meeting as the wing's weekly presenter.
- Evaluated 5 AI assistants (ChatGPT, Gemini, Claude, Copilot and Grok) on a real university scenario using a scoring matrix, budget checks and a retest; delivered an 18-page recommendation report.
- Measured prompt cost in English versus Bangla (360 vs 647 tokens, about 80% more) to inform AI budget planning.
- Completed IT department training in data-to-intelligence workflows, web content writing and prompt engineering.

Capstone Research Team Member – Air Quality Prediction using Machine Learning & AI
East West University, Department of CSE (CSE400C) | 2025 | Supervisor: Dr. Tania Sultana

- Collected 62,188 sensor readings in Badda and Aftabnagar, Dhaka, with a low-cost IoT node (PMSA003, MQ-7, MQ-131, NO_2, DHT22) in a team of three.
- Trained a Random Forest on 7 features to predict the US EPA AQI and compared it with LightGBM, KNN and SVM; reported a time-ordered hold-out score of R^2 0.9933 and RMSE 3.41.
- Exported the model to JSON and shipped it in a real-time dashboard with live Open-Meteo data, a 24-hour forecast and English / Bangla health advice: airqualitypredictionapp.netlify.app

PROJECTS

DocuSense – Grounded RAG Service
Python, FastAPI, FAISS, Claude API, Docker, GitHub Actions | github.com/Majortarif/docusense-rag

- Built a FastAPI microservice that answers questions only from internal documentation and returns a fixed fallback sentence when the answer is not there.
- Implemented markdown-aware chunking, local bge-small-en-v1.5 embeddings and exact cosine search in FAISS, with a similarity threshold calibrated on a golden-set evaluation.
- Added guardrails (prompt-injection screen, similarity gate, citation and numeric-grounding checks), 106 automated tests, Docker and CI.

AgriCycle – Offline-first AgriTech PWA (CSE495)
JavaScript, PWA, service worker, Chart.js, Bangla / English i18n | greencycle-farm.netlify.app | github.com/Majortarif/AgriCycle

- Built a Bangla-first marketplace that connects smallholder farmers directly with buyers, with listings, search, filters and order tracking.
- Developed a crop-waste recycling matcher that ranks partner units by haversine distance and estimates CO_2 saved, plus a 43-topic rule-based farm advisor designed for a future LLM.
- Shipped an installable, offline-capable PWA covering 64 districts with farmer, buyer and admin roles.

CareerPilot AI – Career & Skill Intelligence Platform
JavaScript (ES modules), Tailwind CSS, Chart.js | github.com/Majortarif/CareerPilot-AI-AI-Powered-Career-Skill-Intelligence-Platform

- Built a 12-page prototype for CV and job-description skill extraction, skill-gap scoring, a 12-week learning roadmap and an application tracker with 7 statuses.
- Implemented deterministic, rule-based recommendation logic and deployed it to GitHub Pages through GitHub Actions.

EcoBin 2.0 – Smart Waste Management Dashboard
JavaScript, Tailwind CSS, Cloudflare | github.com/Majortarif/ecobin-smart-waste-dashboard

- Led the smart-bin concept from problem definition to prototype, and built a dashboard monitoring 48 simulated bins with waste analytics and a rule-based route-optimization simulator.

Pulse – Task & Habit Tracker
JavaScript, Tailwind CSS, Netlify | ai-task-habit-tracker.netlify.app

- Built task management with priorities and filters, habit streaks computed from real completion dates, productivity analytics and a rule-based coach.

Digital Phonebook 2026
C, file I/O | github.com/Majortarif/Digital_phone_book

- Rebuilt a 2022 console app with contact CRUD, search across six fields, blocking, favourites, statistics and text-file export.

EDUCATION

B.Sc. in Computer Science and Engineering
East West University, Dhaka | Graduated April 2026 | CGPA 3.01 / 4.00

- Capstone (CSE400C): Air Quality Prediction using Machine Learning & AI
- Course project (CSE495): AgriCycle, a farmer–buyer marketplace and crop-waste recycling app

Higher Secondary Certificate (HSC)
Abdul Kadir Mollah City College | GPA 5.00 / 5.00

Secondary School Certificate (SSC)
Ashuganj Tap Bidyut Kendra High School | GPA 5.00 / 5.00

LEADERSHIP & VOLUNTEERING

- **Volunteer, EWU Robotics Club** (2 years): assisted in organizing technical events and managed team logistics.
- **Volunteer, EWU Science Club** (2 years): supported scientific innovation activities and event coordination.
- **Volunteer, EWU Telecommunication Club** (2 years): took part in workshops and managed communication-based project outreach.

LANGUAGES & ACTIVITIES

- **Languages:** Bengali (native), English (professional working proficiency)
- **Workshops & training:** IoT workshop; CSE 360 Computer Architecture field tour to the National Museum of Science and Technology (2024); YSSE IT department training (2026)
- **Sports:** EWU 5KM Run 2.0 and 3.0, Pantonix Tiger Run Dhaka 2026, Haier Bangladesh Summer Run 2026; university football

REFERENCES

- **Sadia Nur Amin**, Senior Lecturer, Department of Computer Science & Engineering, East West University, Dhaka
- **Dr. Sarwar Jahan**, Associate Professor, Department of Computer Science & Engineering, East West University, Dhaka
- **Dr. Raihan Ul Islam**, Associate Professor, Department of Computer Science & Engineering, East West University, Dhaka